

LINE-BY-LINE COMMENTARY AND SUBMISSION AUDIT FOR THE MANUSCRIPT “Koopman Invariance of Gaussian RKHSs for Linear Stochastic Differential Equations”

ABSTRACT. This document goes through `paper.tex` (1021 source lines, 19 pages when compiled) in source order. For each block of source lines it states what the lines say and derives every mathematical claim with the intermediate steps written out. It also records how each claim was checked and notes caveats. Section 2 summarizes the audit, including the corrections made during it. Section 14 answers whether the manuscript is ready for submission and to which journal it should go.

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Date: 24 September 2026.

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1. HOW TO READ THIS DOCUMENT

Line numbers refer to `paper.tex` as committed together with this commentary. Each block of lines is discussed under up to four headings:

- *Content*: what the lines say;
- *Why it holds*: a derivation, with all intermediate steps;
- *Verified*: how the claim was checked;
- *Notes*: caveats, and corrections made during this audit.

The verification tags are the following.

- A Re-derived by hand. The derivation is written out in this document.
- S Confirmed in exact arithmetic by `verify_symbolic.py` (SymPy, 52 checks in sections S1–S5 of its output).
- N Confirmed in floating-point arithmetic by `verify.py` (38 checks) or `make_figure.py`. The item of Appendix A of the paper is named.
- L Compared with the cited source; the location is named. For [PSWPN] the source is the arXiv PDF supplied with the task, *arXiv:2405.14429v1*, 23 May 2024. Quotations from it are verbatim, except that its mathematical notation has been re-typeset.
- B Build checks: `pdflatex` (TeX Live 2023) and a KaTeX parse of all 960 mathematical snippets of `paper.tex`.

The notation is that of the paper. We write $P \leq Q$ if $Q - P$ is positive semidefinite (PSD), $|x|$ for the Euclidean norm and $\|M\|$ for the spectral norm. The Fourier transform is $\widehat{g}(\xi) = \int g(x)e^{-i\xi \cdot x} dx$, and

$$w_C(\xi) = e^{\frac{1}{4}\xi^\top C^2 \xi}, \quad \kappa_C = \pi^{d/2} \det C,$$

$$D_t = C^2 + 4\Sigma(t) - e^{At}C^2e^{A^\top t}, \quad \Lambda_C = 4BB^\top - (AC^2 + C^2A^\top).$$

Further, $C_t^2 = e^{-At}(C^2 + 2\Sigma(t))e^{-A^\top t}$ and $\tau_t = \det C / \det(C^2 + 2\Sigma(t))^{1/2}$ are the objects of [PSWPN], while $\widetilde{C}_t^2 = e^{-At}(C^2 + 4\Sigma(t))e^{-A^\top t}$ and $\widetilde{\tau}_t = \det C / \det(C^2 + 4\Sigma(t))^{1/2}$ are those of the paper. For Hurwitz A we put $P_C = C^2 + 4\Sigma$. Equation, theorem and section numbers such as “(3.4)” or “Theorem 4.2” refer to the paper; the sections of this commentary are cited as “Section 2” and so on.

2. SUMMARY OF THE AUDIT

2.1. Result. I found no mathematical error in any statement or proof of the manuscript. Every displayed formula and every proof step was re-derived; the derivations are in Sections 3–11. The following checks were run on 24 September 2026.

Check	Result
<code>verify.py</code> (Appendix A, items 1–6 and 8–10)	38 of 38 checks pass; the output is byte-identical to the committed <code>verify_output.txt</code> .
<code>make_figure.py</code> (Figure 1, item 7)	The interval property and the tail bound hold for all 320 values of β . The printed results are identical to <code>figure_output.txt</code> , and the regenerated figure differs from the committed one only in its creation date.
<code>verify_symbolic.py</code> (new)	52 of 52 exact checks pass.
KaTeX parse of <code>paper.tex</code>	960 mathematical snippets, 0 errors.
<code>pdflatex</code> , two passes	0 errors, 0 warnings, 0 overfull or underfull boxes, 19 pages.
Citations of [PSWPN]	Every numbered reference matches arXiv v1: Theorem 2.1 and its proof, (2.3), Remark 2.2(c), Corollary 2.3, Lemma 3.1, Proposition 3.3, (4.1), Proposition 4.1 and its proof, Remark 4.2.

2.2. What is proved, what is computed. All theorems, lemmas, propositions and corollaries are proved analytically. The same holds for every statement in Examples 6.1 and 6.2, with two exceptions: the curve $t_1(\beta)$, and the statement that $\{t \in (0, 40] : D_t \geq 0\}$ is an interval for each of the 320 sampled values of β . Both are numerical and are labelled as such in the paper. The error magnitudes and the Monte Carlo value in Appendix A are numerical. No proof depends on a numerical result.

2.3. Corrections made during the audit. The line numbers refer to the revised `paper.tex`.

- C1 *Notation* (lines 51, 89). The Euclidean norm of a vector was written $\|\cdot\|$, although the Notation paragraph (lines 182–186) reserves $\|\cdot\|$ for the spectral norm and $|\cdot|$ for vectors. The abstract now writes the quadratic form $(x - y)^\top C^{-2}(x - y)$, and line 89 uses $|\cdot|$.
- C2 *Attribution* (lines 150–153, 758–764, 788–789, 889–899). The formula $K^t k_z^C = \tau_t k_{e^{-At}z}^{C_t}$ is established in the *proof* of [PSWPN, Proposition 4.1]; the statement only bounds the norm of $K^t: H_C \rightarrow H_{C_t}$ by $\tau_t^{1/2}$. All citations now say so, and Remark 7.2 derives the formula from (3.2) and (2.3). The paper is therefore self-contained on this point.
- C3 *Self-containedness* (lines 518–520). The proof of Corollary 4.3 quoted the derivative $\dot{f}_x(0)$ from [PSWPN]. It now derives it in one line.
- C4 *An unproved claim* (lines 157–160 and 765–769). Remark 7.2 said that non-invariance is witnessed “only” by functions whose Fourier transforms decay slowly, and this was not proved. It is replaced by

the statement that is proved: a finite kernel expansion f satisfies $|\hat{f}|^2_{w_C} \leq c w_C^{-1}$, so it is never a witness once $C^2 + D_t > 0$. The introduction was adjusted accordingly.

- C5 *A missing hypothesis* (lines 636–637). Remark 5.2 spoke of “the ellipsoid of Σ ”, but Theorem 5.1 does not assume controllability, so Σ may be singular. The remark now says “whenever $\Sigma > 0$ ”.
- C6 *Imprecise explanation* (lines 851–855). Example 7.6 attributed condition (1.5) to “Gaussian kernels”. It now names the mechanism: the Gaussian decay of the spectral density competes with the factor $\exp(-\eta^\top \Sigma(t) \eta)$. It also cites the related Sobolev results [6, 3].
- C7 *Ambiguity* (lines 58–60). In the abstract, “the criterion says” could refer to the fixed-time criterion. The sentence now refers explicitly to invariance for all $t \geq 0$.
- C8 *Precision* (lines 127–128 and 344–345). The phrase “for the bounded continuous functions in H_C ” suggested that only some elements of H_C are bounded and continuous; in fact all are. Also, “bounded and continuous, since $|f(x)| \leq \|f\|_C$ ” justified only boundedness; continuity is now attributed to Lemma 2.1(b).
- C9 *Precision* (lines 142–143). The Fourier membership criterion in “Idea of the proof” now states its hypotheses: f continuous, real-valued and in L^2 .
- C10 *A wrong norm* (lines 873–885 and 889–892). Appendix A, item 2 wrote $\|t^{-1}D_t - \Lambda_C\|$, which by the Notation paragraph is the spectral norm, whereas `verify.py` measures the largest entry. The appendix now defines $|M|_{\max} = \max_{i,j} |M_{ij}|$ and states items 1, 2 and 4 in this norm.
- C11 *Completeness* (lines 871–873 and 920–925). Appendix A now describes the third, exact-arithmetic script.
- C12 *Literature* (lines 77, 797–798, 854–855 and the bibliography). The audit added:
 - Hertel, Philipp, Schaller and Worthmann [3], a follow-up of [PSWPN] on stochastic kernel EDMD;
 - Ikeda, Ishikawa and Sawano [4], on composition operators on RKHSs of analytic kernels.

Köhne et al. was updated from preprint to *SIAM J. Appl. Dyn. Syst.* 24 (2025). Mauroy et al. gained its subtitle and series volume. A note records that numbered statements of [PSWPN] follow arXiv v1.
- C13 *Dead code* (line 10). The hyphenation exception `Mat-ern` was removed. It can never match the word “Matérn”, because the accent command interrupts the word.
- C14 *Submission items* (lines 42–43, 927–935, 938–939). The audit added a commented AI-use statement and a reminder next to the commented author block. The data availability sentence now reads “No external data were used”, which is accurate; before, it said that no data were generated.

C15 *Style* (lines 70–71, 91). The first sentence of the introduction was split for grammar, and “(real) RKHS” is now explicit.

2.4. Novelty check. Web searches on 24 September 2026 found no publication that answers the question of [PSWPN, Remark 2.2(c), Remark 4.2] [19, 20]. According to the search engine’s summary, the July 2026 survey arXiv:2607.01819 still describes the question as open. I could not open the PDF, because the network policy of this environment blocks arxiv.org; the conclusion rests on search results, not on reading the survey. The closest works are:

- [3], on Sobolev spaces for stochastic dynamics, via conditions on a Radon–Nikodym density;
- [6], on Wendland (Sobolev) native spaces for deterministic dynamics;
- [4], on composition operators.

None of them treats Gaussian RKHSs of linear SDEs. A search cannot prove a negative. Before submission the author should repeat the search on arXiv and MathSciNet/zblMATH, which were not reachable from here.

3. FRONT MATTER (LINES 1–64)

Lines 1–10.

Content. Document class `amsart` at 11pt. The packages are `amssymb` (blackboard bold), `amsthm` (theorem environments), `mathtools` (a superset of `amsmath`: `align*`, `gather*`, `smallmatrix`), `graphicx` (Figure 1) and `hyperref` (links). The settings work as follows.

- `\numberwithin` numbers equations as (section.number).
- `\allowdisplaybreaks[1]` permits page breaks inside multi-line displays.
- `\raggedbottom` lets pages end at their natural height.
- `\emergencystretch` adds a third line-breaking pass with extra stretch.
- `\hyphenation` lists exceptions for proper names.

Verified. B: the log has no overfull or underfull boxes; the last three settings were introduced to achieve this.

Notes. Journals reformat with their own class (Section 14).

Lines 12–19.

Content. Theorem-like environments that share one counter numbered within sections. So Theorem 4.1, Theorem 4.2, Corollary 4.3, Theorem 4.4, Remark 4.5 and Corollary 4.6 are consecutive. Examples use the definition style (upright text), and remarks use the remark style.

Lines 21–30.

Content. The macros are the following.

- `\R` is $\mathbb{R}$, `\Sph` is $\mathbb{S}$ (the sphere $\mathbb{S}^{d-1}$), `\Tr` is the trace.
- `\dd` is the differential with a thin space.
- `\FC{C}` is the space $\mathcal{F}_C$ of Fourier transforms.
- `\Ct` is $\tilde{C}_t$ and `\taut` is $\tilde{\tau}_t$.
- `\ip{f}{g}` is $\langle f, g \rangle$, `\hatf` is the wide hat, and `\tp` is the transpose sign.

Verified. B: KaTeX parses all 960 mathematical snippets with these macro definitions.

Lines 32–37.

Content. Title and short title. `amsart` sets titles and running heads in capitals with `\uppercasenonmath`, which does not change mathematics. Writing the plural as `RKHS$\text{s}$` therefore keeps it as “RKHSs” instead of “RKHSS”.

Lines 38–43.

Content. The author block is commented out and must be completed by the submitting author. Lines 42–43 remind the author to activate the AI-use statement (lines 927–935).

Notes. This is a blocking item for submission (Section 14).

Lines 45–47.

Content. MSC 2020 classification and keywords. Primary codes:

- 46E22, Hilbert spaces with reproducing kernels;
- 47D07, Markov semigroups and applications to diffusion processes.

Secondary codes:

- 60H10, stochastic ordinary differential equations;
- 47B33, linear composition operators;
- 42B10, Fourier and Fourier–Stieltjes transforms and other transforms of Fourier type.

Notes. The code 47B32 (linear operators in reproducing-kernel Hilbert spaces) would also fit. Adding it is optional.

Lines 49–62.

Content. The abstract. Sentence by sentence:

- (1) The result of [PSWPN]: sufficiency of $\frac{1}{2}(AC^2 + C^2A^\top) \leq BB^\top$ under “A Hurwitz, (A, B) controllable”. L: [PSWPN, Theorem 2.1] and the standing assumptions of [PSWPN, Section 2].
- (2) “They asked whether this condition is also necessary.” L: [PSWPN, Remark 2.2(c)] reads “It is not clear whether the condition (2.3) is also necessary for Koopman invariance of H_C .” [PSWPN, Remark 4.2] reads “It is not clear whether the statement of Theorem 2.1 is actually an equivalence.”

- (3) The answer and the sharp condition: Theorem 1.1(b), which is Theorem 4.4, and Corollary 1.2.
- (4) “ K^t maps H_C isomorphically onto the Gaussian RKHS with matrix $\tilde{C}_t$ ”: Theorem 4.2. The map is $\tilde{\tau}_t^{1/2}$ times a unitary operator.
- (5) The fixed-time criterion and the norm: Corollary 4.3 and Theorem 4.1.
- (6) The Hurwitz case: Theorem 5.1(b).
- (7) The method: Lemmas 2.1 and 3.2.

Notes. Corrections C1 and C7 apply here.

4. PAPER SECTION 1: INTRODUCTION (LINES 67–186)

Lines 70–79.

Content. The Koopman operator of a stochastic system is $(K^t f)(x) = \mathbb{E}[f(X_t) | X_0 = x]$. It is linear, it can be learned from data (EDMD and its kernel variants), and error bounds for kernel methods profit from the invariance of the RKHS.

Why it holds. Linearity: $K^t(\alpha f + \beta g) = \alpha K^t f + \beta K^t g$ by linearity of conditional expectation, whatever the dynamics.

Verified. L: the bibliographic data are checked in Section 12. The remaining statements are descriptive.

Lines 81–103.

Content. The setting of [PSWPN].

- The SDE (1.1).
- The kernel k^C and its real RKHS H_C .
- The conditional covariance $\Sigma(t)$ in (1.2).
- The standing assumptions: $m \leq d$, (A, B) controllable, A Hurwitz.
- The sufficient condition (1.3), under which $\|K^t f\|_C \leq e^{-t \operatorname{Tr} A/2} \|f\|_C$.

The open question is whether (1.3) is also necessary.

Why it holds. That $\Sigma(t)$ is the covariance of X_t given X_0 is shown in Section 6 (lines 336–343).

Verified. L. [PSWPN, Section 2] assumes “ $m \leq d$ ”, “ (A, B) is controllable and A is Hurwitz”. [PSWPN, Theorem 2.1] reads: “Let $C \in \mathbb{R}^{d \times d}$ be a symmetric positive definite matrix such that $\frac{1}{2}(AC^2 + C^2 A^\top) \leq BB^\top$. (2.3) Then the RKHS H_C is invariant under each Koopman operator K^t , $t \geq 0$, associated with the SDE (2.1), and we have $\|K^t f\|_C \leq e^{t \operatorname{Tr}(-A)/2} \cdot \|f\|_C$.” Both remarks cited at line 103 are quoted in Section 3.

Notes. Corrections C1 and C15 apply here.

Lines 105–125.

Content. Theorem 1.1. For arbitrary A and B and symmetric positive definite C :

- (a) $K^t H_C \subset H_C$ if and only if $e^{At} C^2 e^{A^\top t} \leq C^2 + 4\Sigma(t)$, and then $\|K^t\| = e^{-t \operatorname{Tr} A/2}$;
- (b) invariance for all $t \geq 0$ holds if and only if $\frac{1}{2}(AC^2 + C^2 A^\top) \leq 2BB^\top$.
If this fails, invariance fails on a whole interval $(0, \varepsilon)$.

Why it holds. (a) is Theorem 4.1(a),(b), because $D_t \geq 0$ is (1.4). (b) is Theorem 4.4, and the “interval” statement is proved at lines 558–563 (Section 7).

Notes.

- No standing assumption is needed: A may be unstable, B may vanish, and m is arbitrary.
- Part (a) shows that the bound of [PSWPN, Theorem 2.1] is *sharp*. Whenever H_C is invariant, $\sup_{f \neq 0} \|K^t f\|_C / \|f\|_C = e^{-t \operatorname{Tr} A/2}$. For Hurwitz A this number exceeds 1 and grows exponentially in t .

Lines 127–129.

Content. $K^t f$ is defined pointwise, which is meaningful because every $f \in H_C$ is bounded and continuous. Under the assumptions of [PSWPN] this agrees with the $L^2(\mu)$ definition.

Why it holds. See lines 344–345 and Remark 3.1 (Section 6).

Notes. Correction C8 applies here.

Lines 131–139.

Content. Corollary 1.2. Condition (1.3) implies (1.5) but not conversely. The matrices $A = \begin{pmatrix} -1 & 5 \\ 0 & -1 \end{pmatrix}$ and $B = C = I$ separate the two conditions. Moreover, (1.5) is (1.3) with $C/\sqrt{2}$ in place of C .

Why it holds. See the proof at lines 565–572 (Section 7). The last statement holds because

$$\frac{1}{2}(A(C/\sqrt{2})^2 + (C/\sqrt{2})^2 A^\top) = \frac{1}{4}(AC^2 + C^2 A^\top),$$

so “ $\leq BB^\top$ ” for $C/\sqrt{2}$ is the same as “ $\frac{1}{2}(AC^2 + C^2 A^\top) \leq 2BB^\top$ ” for C .

Verified. S: S5 (the identity) and S3 (eigenvalues $\frac{3}{2}$ and $-\frac{7}{2}$ for $\beta = 5$). N: item 6 ($\beta = 5$).

Notes. The following observation is not in the paper. Since

$$k^{C/\sqrt{2}}(x, y) = \exp(-2(x - y)^\top C^{-2}(x - y)) = k^C(x, y)^2,$$

the sharp condition for H_C is exactly the sufficient condition of [PSWPN] for the *squared* kernel $(k^C)^2$. It could be added as a one-line remark.

Lines 141–146.

Content. Idea of the proof: a Fourier description of H_C ; the Fourier form of K^t ; comparison of two Gaussian weights; differentiation at $t = 0$.

Why it holds. Lemma 2.1, Lemma 3.2, Theorem 4.1 and Theorem 4.4.

Notes. Correction C9 applies here.

Lines 148–160.

Content. “Where the factor two comes from.” [PSWPN] track kernel sections. Averaging a Gaussian bump over the Gaussian transition law adds $\Sigma(t)$ to its covariance $\frac{1}{2}C^2$, and this gives $K^t k_z^C = \tau_t k_{e^{-At}z}^{C_t}$. Membership in H_C is decided by $|\widehat{f}|^2$, where the smoothing factor appears squared. This turns $2\Sigma(t)$ into $4\Sigma(t)$.

Why it holds. Kernel sections. By (2.3), $\widehat{k_z^C}(\eta) = \kappa_C e^{-i\eta \cdot z} w_C(\eta)^{-1}$. By (3.2) with $\eta = e^{-A^\top t} \xi$, and since $(e^{-A^\top t} \xi) \cdot z = \xi \cdot e^{-At} z$,

$$\begin{aligned} \widehat{K^t k_z^C}(\xi) &= e^{-t \operatorname{Tr} A} \kappa_C e^{-i\xi \cdot e^{-At} z} \exp\left(-\frac{1}{4} \xi^\top e^{-At} C^2 e^{-A^\top t} \xi\right) \\ &\quad \times \exp\left(-\frac{1}{2} \xi^\top e^{-At} \Sigma(t) e^{-A^\top t} \xi\right) \\ &= e^{-t \operatorname{Tr} A} \kappa_{C_t} e^{-i\xi \cdot e^{-At} z} w_{C_t}(\xi)^{-1}. \end{aligned}$$

Also $\widehat{k_{e^{-At}z}^{C_t}}(\xi) = \kappa_{C_t} e^{-i\xi \cdot e^{-At} z} w_{C_t}(\xi)^{-1}$. The two transforms therefore differ by the factor

$$\frac{e^{-t \operatorname{Tr} A} \kappa_C}{\kappa_{C_t}} = \frac{e^{-t \operatorname{Tr} A} \det C}{\det C_t} = \frac{e^{-t \operatorname{Tr} A} \det C}{e^{-t \operatorname{Tr} A} \det(C^2 + 2\Sigma(t))^{1/2}} = \tau_t,$$

where $\det C_t = (\det e^{-At}) \det(C^2 + 2\Sigma(t))^{1/2} = e^{-t \operatorname{Tr} A} \det(C^2 + 2\Sigma(t))^{1/2}$. Both sides are continuous L^2 functions with the same Fourier transform, hence $K^t k_z^C = \tau_t k_{e^{-At}z}^{C_t}$.

Squared factor. For general $f \in H_C$, (3.2) gives

$$|\widehat{K^t f}(\xi)|^2 = e^{-2t \operatorname{Tr} A} |\widehat{f}(e^{-A^\top t} \xi)|^2 \exp(-\xi^\top e^{-At} \Sigma(t) e^{-A^\top t} \xi).$$

The smoothing factor now carries $\Sigma(t)$, not $\frac{1}{2}\Sigma(t)$. Let C' be a candidate matrix and substitute $\xi = e^{A^\top t} \eta$. Then the weight $\exp(-\eta^\top \Sigma(t) \eta + \frac{1}{4} \eta^\top e^{At} C'^2 e^{A^\top t} \eta)$ must be compared with $w_C(\eta)$. The two are equal exactly when $e^{At} C'^2 e^{A^\top t} = C^2 + 4\Sigma(t)$, i.e. for $C' = \tilde{C}_t$ (Theorem 4.2).

Loss in [PSWPN]. Since $\tilde{C}_t^2 - C_t^2 = 2e^{-At} \Sigma(t) e^{-A^\top t} \geq 0$, the range $H_{\tilde{C}_t}$ is a subspace of H_{C_t} , and a proper one when $\Sigma(t) \neq 0$ (Corollary 4.3). Checking $H_{C_t} \subset H_C$ asks for more than invariance needs.

Verified. L. The formula appears in the proof of [PSWPN, Proposition 4.1], whose last display ends with “hence, $K^t k_z^C = \tau_t \cdot k_{e^{-At}z}^{C_t} \in H_{C_t}$ ”. The statement of [PSWPN, Proposition 4.1] says that K^t “maps H_C boundedly into H_{C_t} with norm not exceeding $\tau_t^{1/2}$ ”. The equivalence “(1.3) $\iff H_{C_t} \subset H_C$ for all t ” is in the proof of [PSWPN, Theorem 2.1]: “We prove that $C_t^2 \geq C^2$ for all $t \geq 0$ if and only if (2.3) is satisfied”, combined with [PSWPN, Proposition 3.3]. S: S2 checks the kernel-section formula by an x -space Gaussian convolution for $d = 1$. N: item 5 is a Monte Carlo check.

Notes. Corrections C2 and C4 apply here.

Lines 161–167.

Content. For Hurwitz A , $\Sigma = \lim_{t \rightarrow \infty} \Sigma(t)$ exists and solves $A\Sigma + \Sigma A^\top = -BB^\top$. The two conditions become Lyapunov inequalities for $C^2 + 4\Sigma$ and for $C^2 + 2\Sigma$. Invariance is equivalent to forward invariance of an ellipsoid, and it holds for all large t .

Why it holds. Existence of the limit: if A is Hurwitz, there are $M \geq 1$ and $\alpha > 0$ with $\|e^{As}\| \leq Me^{-\alpha s}$. Hence $\|e^{As}BB^\top e^{A^\top s}\| \leq M^2\|B\|^2 e^{-2\alpha s}$ is integrable on $[0, \infty)$. The Lyapunov equation and the equivalences are Theorem 5.1 (Section 8).

Lines 169–173.

Content. The Fourier argument extends to translation-invariant kernels (Proposition 7.5). Sobolev (Matérn) spaces are always invariant (Example 7.6).

Lines 175–186.

Content. Organization and notation.

Verified. A: the section list matches the section titles at lines 188, 333, 412, 588, 645, 735 and 868.

5. PAPER SECTION 2: GAUSSIAN RKHSs IN FOURIER VARIABLES (LINES 188–330)

Lines 191–197.

Content. Fourier conventions; Plancherel's identity; the inverse transform $F^\vee$; Hermitian functions.

Why it holds. We check the claims in order.

- (1) *Plancherel.* For $g \in L^1 \cap L^2$, $\|\widehat{g}\|_{L^2}^2 = (2\pi)^d \|g\|_{L^2}^2$ with this normalization. Since $L^1 \cap L^2$ is dense in L^2 , the map extends uniquely to L^2 .
- (2) $F^\vee$ is continuous for $F \in L^1$. The integrand $F(\xi)e^{i\xi \cdot x}$ is dominated by $|F(\xi)|$, so continuity in x follows by dominated convergence.
- (3) $F^\vee \in L^2$ and $\widehat{F^\vee} = F$ for $F \in L^1 \cap L^2$. We have $F^\vee(x) = (2\pi)^{-d} \widehat{F}(-x)$. So $F^\vee \in L^2$ by Plancherel, and $\widehat{F^\vee} = F$ is the L^2 inversion theorem.
- (4) $g = (\widehat{g})^\vee$ everywhere if $g \in L^2 \cap C$ and $\widehat{g} \in L^1$. Here $\widehat{g} \in L^1 \cap L^2$, so $(\widehat{g})^\vee$ is continuous and has Fourier transform $\widehat{g}$. By injectivity of the L^2 Fourier transform, $(\widehat{g})^\vee = g$ almost everywhere. The set where two continuous functions differ is open; if it is also null, it is empty.
- (5) g real $\Rightarrow \widehat{g}$ Hermitian. For $g \in L^1$ we have $\widehat{\overline{g(\xi)}} = \int g(x)e^{i\xi \cdot x} dx = \widehat{g}(-\xi)$. For $g \in L^2$, apply this to the truncations $g\mathbf{1}_{\{|x| \leq n\}}$. Their transforms converge in L^2 , hence almost everywhere along a subsequence, and the identity passes to the limit.

(6) $F \in L^1 \cap L^2$ Hermitian $\Rightarrow F^\vee$ real. Using the substitution $\xi \rightarrow -\xi$,

$$\overline{F^\vee(x)} = (2\pi)^{-d} \int \overline{F(\xi)} e^{-i\xi \cdot x} d\xi = (2\pi)^{-d} \int F(-\xi) e^{-i\xi \cdot x} d\xi = F^\vee(x).$$

Verified. A.

Lines 198–201.

Content. The Gaussian integral (2.1), for $P > 0$:

$$\int_{\mathbb{R}^d} e^{-x^\top P x - i\xi \cdot x} dx = \pi^{d/2} (\det P)^{-1/2} e^{-\frac{1}{4}\xi^\top P^{-1}\xi}.$$

Why it holds. One dimension. Let $I(\xi) := \int_{\mathbb{R}} e^{-px^2 - i\xi x} dx$ with $p > 0$. Differentiating under the integral (dominated by $|x|e^{-px^2}$) gives $I'(\xi) = \int (-ix) e^{-px^2 - i\xi x} dx$. Since $-ix e^{-px^2} = \frac{i}{2p} \frac{d}{dx} e^{-px^2}$, integration by parts gives

$$I'(\xi) = \frac{i}{2p} \int \left(\frac{d}{dx} e^{-px^2} \right) e^{-i\xi x} dx = -\frac{i}{2p} \int e^{-px^2} (-i\xi) e^{-i\xi x} dx = -\frac{\xi}{2p} I(\xi).$$

Together with $I(0) = \sqrt{\pi/p}$ this gives $I(\xi) = \sqrt{\pi/p} e^{-\xi^2/(4p)}$.

Dimension d . Write $P = Q\Lambda Q^\top$ with Q orthogonal and $\Lambda = \text{diag}(\lambda_j)$, $\lambda_j > 0$. Substitute $x = Qu$ (so $|\det Q| = 1$) and put $\zeta = Q^\top \xi$. Then $x^\top P x = \sum_j \lambda_j u_j^2$ and $\xi \cdot x = \zeta \cdot u$, and the integral factorizes:

$$\begin{aligned} \prod_{j=1}^d \sqrt{\pi/\lambda_j} e^{-\zeta_j^2/(4\lambda_j)} &= \pi^{d/2} (\det P)^{-1/2} e^{-\frac{1}{4}\zeta^\top \Lambda^{-1} \zeta} \\ &= \pi^{d/2} (\det P)^{-1/2} e^{-\frac{1}{4}\xi^\top P^{-1} \xi}. \end{aligned}$$

Verified. A; S: S1 checks $d = 1$.

Lines 203–215.

Content. The definitions (2.2) of w_C and κ_C , the space $\mathcal{F}_C$, the transform of a kernel section (2.3), and the integral (2.4).

Why it holds. (2.3). Substitute $x = y + u$ and apply (2.1) with $P = C^{-2}$. Since $(\det C^{-2})^{-1/2} = \det C$ and $P^{-1} = C^2$,

$$\begin{aligned} \widehat{k_y^C}(\xi) &= e^{-i\xi \cdot y} \int e^{-u^\top C^{-2} u - i\xi \cdot u} du \\ &= e^{-i\xi \cdot y} \pi^{d/2} \det C e^{-\frac{1}{4}\xi^\top C^2 \xi} = \kappa_C e^{-i\xi \cdot y} w_C(\xi)^{-1}. \end{aligned}$$

(2.4). Apply (2.1) with $\xi = 0$ and $P = \frac{1}{4}C^2$. Then $\det P = 4^{-d}(\det C)^2$, so

$$\int w_C^{-1} d\xi = \pi^{d/2} 2^d / \det C = (4\pi)^{d/2} / \det C.$$

Verified. A; S: S1 checks (2.3) and (2.4) for $d = 1$.

Notes. $\mathcal{F}_C$ is to be read modulo null sets, as usual for L^2 -type spaces.

Lines 217–230.

Content. Lemma 2.1:

- (a) $\mathcal{F}_C \subset L^1 \cap L^2$, with an explicit L^1 bound;
- (b) H_C consists of the continuous, real, L^2 functions with $\widehat{f} \in \mathcal{F}_C$, and its inner product is (2.5);
- (c) $f \mapsto \widehat{f}$ is a bijection from H_C onto $\mathcal{F}_C$.

Lines 232–238.

Content. Proof of (a).

Why it holds. By Cauchy–Schwarz and (2.4),

$$\|F\|_{L^1} = \int |F| w_C^{1/2} w_C^{-1/2} \leq \left(\int |F|^2 w_C \right)^{1/2} \left(\frac{(4\pi)^{d/2}}{\det C} \right)^{1/2}.$$

Since $w_C \geq 1$, also $\|F\|_{L^2}^2 \leq \int |F|^2 w_C$.

Verified. A.

Lines 240–246.

Content. $\mathcal{H}$ (the right-hand side of (b)) with the form (2.5) is a real inner product space.

Why it holds. *Vector space:* $|\widehat{f} + \widehat{g}|^2 \leq 2|\widehat{f}|^2 + 2|\widehat{g}|^2$.

Absolute convergence:

$$\int |\widehat{f}||\widehat{g}| w_C \leq \left(\int |\widehat{f}|^2 w_C \right)^{1/2} \left(\int |\widehat{g}|^2 w_C \right)^{1/2}.$$

Realness: put $I(\xi) = \widehat{f}(\xi) \overline{\widehat{g}(\xi)} w_C(\xi)$. Hermitian symmetry and evenness of w_C give $I(-\xi) = \overline{I(\xi)}$, so $\int I = \int \overline{I} = \overline{\int I}$.

Definiteness: $\langle f, f \rangle = 0$ forces $\widehat{f} = 0$ almost everywhere, because $w_C > 0$. Then $f = 0$ in L^2 , and $f = 0$ everywhere by continuity.

Verified. A.

Lines 248–251.

Content. The inversion formula (2.6) holds at every x .

Why it holds. $\widehat{f} \in L^1$ by (a); now apply item 4 of the conventions (lines 191–197).

Lines 253–256.

Content. $\mathcal{H}$ is complete.

Why it holds. Let (f_n) be Cauchy in $\mathcal{H}$. By (2.5), $(\widehat{f_n})$ is Cauchy in $L^2(w_C d\xi)$, which is complete, so $\widehat{f_n} \rightarrow F$ there.

- A subsequence converges almost everywhere. Since $\widehat{f_n}(-\xi) = \overline{\widehat{f_n}(\xi)}$ off a null set, and the reflection of a null set is null, F is Hermitian.

- Thus $F \in \mathcal{F}_C \subset L^1 \cap L^2$, so $f := F^\vee$ is continuous, real, in L^2 , and $\widehat{f} = F$.
- Hence $f \in \mathcal{H}$, and $\|f_n - f\|_{\mathcal{H}} \rightarrow 0$.

Verified. A.

Lines 258–260.

Content. The kernel sections belong to $\mathcal{H}$.

Why it holds. By (2.3), $\widehat{k_y^C}$ is Hermitian: $\widehat{k_y^C}(-\xi) = \kappa_C e^{i\xi \cdot y} w_C(\xi)^{-1} = \overline{\widehat{k_y^C}(\xi)}$. Moreover $\int |\widehat{k_y^C}|^2 w_C = \kappa_C^2 \int w_C^{-1} < \infty$, and k_y^C is continuous, real and square integrable.

Notes. The normalization is consistent:

$$\begin{aligned} \|k_y^C\|^2 &= (2\pi)^{-d} \kappa_C^{-1} \kappa_C^2 \int w_C^{-1} = (2\pi)^{-d} \pi^{d/2} \det C \frac{(4\pi)^{d/2}}{\det C} \\ &= (2\pi)^{-d} (2\pi)^d = 1 = k^C(y, y). \end{aligned}$$

S: S1 checks this for $d = 1$ and for general d .

Lines 262–267.

Content. The reproducing property.

Why it holds. By (2.3), $\widehat{k_y^C}(\xi) = \kappa_C e^{i\xi \cdot y} w_C(\xi)^{-1}$. So

$$\langle f, k_y^C \rangle_{\mathcal{H}} = (2\pi)^{-d} \kappa_C^{-1} \int \widehat{f} \kappa_C e^{i\xi \cdot y} w_C^{-1} w_C d\xi = (2\pi)^{-d} \int \widehat{f}(\xi) e^{i\xi \cdot y} d\xi = f(y),$$

by (2.6).

Verified. A.

Lines 269–279.

Content. $\mathcal{H} = H_C$ isometrically: the uniqueness part of the Moore–Aronszajn theorem.

Why it holds. The argument has four steps.

- *Density of $H_0 = \text{span}\{k_y^C\}$.* In any RKHS, $f \perp H_0$ implies $f(y) = \langle f, k_y^C \rangle = 0$ for all y , so $f = 0$. This applies both to $\mathcal{H}$ and to H_C .
- *Equal norms on H_0 .* By the reproducing property, in both spaces the squared norm of $\sum_j \alpha_j k_{y_j}^C$ equals $\sum_{i,j} \alpha_i \alpha_j k^C(y_i, y_j)$.
- *Pointwise control.* $|h(x)| = |\langle h, k_x^C \rangle| \leq \|h\| \|k_x^C\| = \|h\|$, because $\|k_x^C\|^2 = k^C(x, x) = 1$.
- *Limits.* Let $f \in \mathcal{H}$ and choose $f_n \in H_0$ with $f_n \rightarrow f$ in $\mathcal{H}$. Then (f_n) is Cauchy in H_C and converges there to some g . Pointwise limits coincide, so $g = f$ and the norms agree. The converse inclusion is the same argument with the roles exchanged; it uses the completeness of $\mathcal{H}$ (lines 253–256).

Finally, polarization gives $\langle f, g \rangle = \frac{1}{4}(\|f + g\|^2 - \|f - g\|^2)$ in the real space. *Verified.* A; L: the uniqueness of the RKHS with a given kernel is [1] and [10].

Lines 281–284.

Content. Proof of (c).

Why it holds. *Injective:* $\widehat{f}$ determines f in L^2 , hence everywhere by continuity. *Surjective:* for $F \in \mathcal{F}_C$, the function $F^\vee$ is real, continuous and in L^2 , with $\widehat{F^\vee} = F$, so $F^\vee \in H_C$ by (b).

Lines 286–288.

Content. Lemma 2.1 is the Gaussian case of Wendland’s Fourier description of native spaces [16, Theorem 10.12]. A power-series description is in [13].

Why it holds. Wendland uses $\widehat{f}_W(\omega) = (2\pi)^{-d/2} \int f(x) e^{-i\omega \cdot x} dx$ and $(f, g) = (2\pi)^{-d/2} \int \widehat{f}_W \widehat{g}_W / \widehat{\Phi}_W$. Substituting $\widehat{f}_W = (2\pi)^{-d/2} \widehat{f}$ and $\widehat{\Phi}_W = (2\pi)^{-d/2} \widehat{\varphi}$ gives

$$(2\pi)^{-d/2} \frac{(2\pi)^{-d} \int \widehat{f} \widehat{g} / \widehat{\varphi}}{(2\pi)^{-d/2}} = (2\pi)^{-d} \int \widehat{f} \widehat{g} / \widehat{\varphi}.$$

For $\varphi = k^C(\cdot, 0)$ we have $\widehat{\varphi} = \kappa_C w_C^{-1}$, which gives (2.5).

Notes. The theorem number and the normalization of [16] are quoted from memory of the book; the book was not accessible here. The author should confirm them in the book. Lemma 2.1 is proved independently in the paper, so nothing depends on the citation.

Lines 290–297.

Content. Lemma 2.2 (exponential multipliers). Let S be symmetric. Then $\int |F|^2 w_C e^{\frac{1}{4} \xi^\top S \xi} < \infty$ for all $F \in \mathcal{F}_C$ if and only if $S \leq 0$. Otherwise a real, even witness supported in a double cone exists.

Lines 299–316.

Content. The proof.

Why it holds. If $S \leq 0$, the multiplier is at most 1. Otherwise, let $\lambda > 0$ be an eigenvalue with unit eigenvector v and put $U = \{\theta \in \mathbb{S}^{d-1} : \theta^\top S \theta > \lambda/2\}$.

- U is relatively open, since $\theta \mapsto \theta^\top S \theta$ is continuous. It is symmetric and contains v , since $v^\top S v = \lambda$.
- The cone $\Gamma = \{r\theta : r > 0, \theta \in U\}$ is the preimage of U under the continuous map $\xi \mapsto \xi/|\xi|$ on $\mathbb{R}^d \setminus \{0\}$, hence open. On Γ we have $\xi^\top S \xi > \frac{\lambda}{2} |\xi|^2$.
- $F_S = w_C^{-1/2} (1 + |\xi|^2)^{-d/2} \mathbf{1}_\Gamma$ is real and even, hence Hermitian. Moreover $\int |F_S|^2 w_C = \int_\Gamma (1 + |\xi|^2)^{-d} < \infty$, because $r^{d-1} (1 + r^2)^{-d}$ is bounded near 0 and $O(r^{-d-1})$ at ∞ .

Multiplying by $e^{\frac{1}{4}\xi^\top S\xi} \geq e^{\lambda|\xi|^2/8}$ on Γ (note $\frac{1}{4} \cdot \frac{\lambda}{2} = \frac{\lambda}{8}$) gives $|U| \int_0^\infty r^{d-1}(1+r^2)^{-d} e^{\lambda r^2/8} dr = \infty$. Here $|U| > 0$ because U is a nonempty open subset of the sphere. For $d = 1$, $U = \{-1, 1\}$ with counting measure.

Verified. A.

Notes. The witness is chosen so that $\int |F_S|^2 w_C$ converges only because of a polynomial factor. Any Gaussian growth in a cone therefore destroys integrability. This is the precise sense of “decays as slowly as membership permits”.

Lines 318–330.

Content. Corollary 2.3: $H_{C_1} \subset H_{C_2}$ if and only if $C_1^2 \geq C_2^2$, and then $\|f\|_{C_2} \leq (\det C_1 / \det C_2)^{1/2} \|f\|_{C_1}$.

Why it holds. By Lemma 2.1(b),(c), inclusion means $\int |F|^2 w_{C_2} < \infty$ for all $F \in \mathcal{F}_{C_1}$. Write $w_{C_2} = w_{C_1} e^{\frac{1}{4}\xi^\top (C_2^2 - C_1^2)\xi}$ and apply Lemma 2.2 with $S = C_2^2 - C_1^2$. For the norm, use $w_{C_2} \leq w_{C_1}$ in (2.5):

$$\|f\|_{C_2}^2 \leq \frac{\kappa_{C_1}}{\kappa_{C_2}} \|f\|_{C_1}^2 = \frac{\det C_1}{\det C_2} \|f\|_{C_1}^2.$$

Verified. A; L: [PSWPN, Proposition 3.3] states “ $H_{C_1} \hookrightarrow H_{C_2}$ if and only if $C_1^2 \geq C_2^2$. In this case, we have $\|f\|_{C_2} \leq (\det C_1 / \det C_2)^{1/2} \|f\|_{C_1}$ ”. The statement and the constant are identical.

6. PAPER SECTION 3: THE KOOPMAN OPERATORS IN FOURIER VARIABLES (LINES 333–409)

Lines 336–343.

Content. The solution is $X_t = e^{At}x + Z_t$ with $Z_t \sim \mathcal{N}(0, \Sigma(t))$, and hence (3.1): $(K^t f)(x) = \int f(e^{At}x + y) \gamma_t(dy)$.

Why it holds. Variation of constants. By Itô’s formula, $d(e^{-At}X_t) = -Ae^{-At}X_t dt + e^{-At}dX_t = e^{-At}B dW_t$. So $X_t = e^{At}x + \int_0^t e^{A(t-s)}B dW_s$.

Law of Z_t . Z_t is a Wiener integral of a deterministic integrand, hence centered Gaussian. By the Itô isometry and the substitution $r = t - s$,

$$\mathbb{E}[Z_t Z_t^\top] = \int_0^t e^{A(t-s)} B B^\top e^{A^\top(t-s)} ds = \int_0^t e^{Ar} B B^\top e^{A^\top r} dr = \Sigma(t).$$

Formula (3.1) is the expectation of $f(e^{At}x + Z_t)$.

Verified. A; L: [15] (distribution of the multivariate Ornstein–Uhlenbeck process). N: items 1 and 5.

Lines 344–345.

Content. Every $f \in H_C$ is continuous (Lemma 2.1(b)) and bounded by $\|f\|_C$.

Why it holds. $|f(x)| = |\langle f, k_x^C \rangle| \leq \|f\|_C \|k_x^C\|_C = \|f\|_C$.

Notes. Correction C8 applies here.

Lines 347–353.

Content. Remark 3.1. The $L^2(\mu)$ reading of $K^t H_C \subset H_C$ in [PSWPN] agrees with the pointwise reading.

Why it holds. Under controllability, $\Sigma > 0$, so $\mu = \mathcal{N}(0, \Sigma)$ has an everywhere positive density. Two continuous functions that agree μ -almost everywhere agree Lebesgue-almost everywhere, hence everywhere. The $L^2(\mu)$ Koopman operator is defined by the same conditional expectation, so its class of $K^t f$ contains the continuous function (3.1).

Verified. L: [PSWPN, Lemma 3.1] reads “the RKHS H_C is densely and continuously embedded in $L^2(\mu)$ ”. [PSWPN, Section 2] reads “ Σ is positive definite thanks to controllability of (A, B) ”.

Lines 355–362.

Content. Lemma 3.2. For bounded, continuous $f \in L^2$, $K^t f$ is bounded, continuous and in L^2 , and (3.2) holds:

$$\widehat{K^t f}(\xi) = e^{-t \operatorname{Tr} A} \widehat{f}(e^{-A^\top t} \xi) \exp\left(-\frac{1}{2} \xi^\top e^{-At} \Sigma(t) e^{-A^\top t} \xi\right).$$

Lines 364–380.

Content. The proof: a convolution step and a dilation step.

Why it holds. Continuity: by dominated convergence in (3.1), with dominating constant $\sup |f|$.

Convolution. $J_t g(z) = \int g(z + y) \gamma_t(dy)$. Minkowski’s integral inequality applied to $|g|$ gives $\|\int |g(\cdot + y)| \gamma_t(dy)\|_{L^2} \leq \int \|g(\cdot + y)\|_{L^2} \gamma_t(dy) = \|g\|_{L^2}$. Hence the integral converges absolutely for almost every z , and $\|J_t g\|_{L^2} \leq \|g\|_{L^2}$. For $g \in L^1 \cap L^2$, Fubini’s theorem applies, because $\iint |g(z + y)| dz \gamma_t(dy) = \|g\|_{L^1}$. With $u = z + y$,

$$\int g(z + y) e^{-i\xi \cdot z} dz = e^{i\xi \cdot y} \widehat{g}(\xi), \quad \int e^{i\xi \cdot y} \gamma_t(dy) = e^{-\frac{1}{2} \xi^\top \Sigma(t) \xi}.$$

The second identity is the characteristic function of $\mathcal{N}(0, \Sigma(t))$; it holds also for singular $\Sigma(t)$, which is how degenerate Gaussian measures are defined. Both sides of $\widehat{J_t g} = \widehat{g} e^{-\frac{1}{2} \xi^\top \Sigma \xi}$ are bounded linear maps $L^2 \rightarrow L^2$, since the multiplier is at most 1. So the identity extends by density.

Dilation. For $u \in L^1 \cap L^2$ and invertible E , substitute $z = Ex$, $dx = |\det E|^{-1} dz$, and use $\xi \cdot E^{-1} z = (E^{-\top} \xi) \cdot z$:

$$\widehat{u \circ E}(\xi) = \int u(Ex) e^{-i\xi \cdot x} dx = |\det E|^{-1} \widehat{u}(E^{-\top} \xi).$$

Both sides are bounded on L^2 : $\|u \circ E\|_{L^2} = |\det E|^{-1/2} \|u\|_{L^2}$. So the formula extends by density.

Assembly. Take $E = e^{At}$. Jacobi's formula gives $\det e^{At} = e^{t \operatorname{Tr} A} > 0$, and $E^{-\top} = e^{-A^\top t}$. For bounded f , $J_t f$ is defined everywhere and agrees with the L^2 version almost everywhere. Then $K^t f = (J_t f) \circ e^{At}$ yields (3.2).

Verified. A; S: S2 checks the characteristic function and (3.2) for $f = k_z^C$, $d = 1$. N: items 4, 5 and 9 use consequences of (3.2).

Lines 382–388.

Content. The definitions (3.3) of D_t and Λ_C , and three rewritings.

Why it holds. (1.4) is $D_t \geq 0$ by definition. (1.5) is equivalent to $AC^2 + C^2 A^\top \leq 4BB^\top$, that is $\Lambda_C \geq 0$. Multiplying (1.3) by 2 gives $2BB^\top - (AC^2 + C^2 A^\top) \geq 0$.

Lines 390–409.

Content. Proposition 3.3, the norm identity (3.4), valid in $[0, \infty]$:

$$\int |\widehat{K^t f}|^2 w_C d\xi = e^{-t \operatorname{Tr} A} \int |\widehat{f}(\eta)|^2 w_C(\eta) e^{-\frac{1}{4}\eta^\top D_t \eta} d\eta.$$

Why it holds. Square (3.2) and multiply by $w_C(\xi)$ to get

$$e^{-2t \operatorname{Tr} A} |\widehat{f}(e^{-A^\top t} \xi)|^2 \exp(-\xi^\top e^{-At} \Sigma e^{-A^\top t} \xi + \frac{1}{4} \xi^\top C^2 \xi).$$

Substitute $\xi = e^{A^\top t} \eta$, with $d\xi = |\det e^{A^\top t}| d\eta = e^{t \operatorname{Tr} A} d\eta$. The exponent becomes

$$-\eta^\top \Sigma(t) \eta + \frac{1}{4} \eta^\top e^{At} C^2 e^{A^\top t} \eta = \frac{1}{4} \eta^\top C^2 \eta - \frac{1}{4} \eta^\top (C^2 + 4\Sigma(t) - e^{At} C^2 e^{A^\top t}) \eta.$$

The substitution is valid for nonnegative integrands with values in $[0, \infty]$.

Verified. A; S: S2 checks that the integrands agree after the substitution ($d = 1$, $f = k_z^C$). N: item 4 (the value of $\|K^t k_z^C\|_C^2$).

7. PAPER SECTION 4: PROOFS OF THEOREM 1.1 AND COROLLARY 1.2 (LINES 412–585)

Lines 415–426.

Content. Theorem 4.1 (fixed time):

- (a) $K^t H_C \subset H_C \iff D_t \geq 0$;
- (b) if $D_t \geq 0$, then $\|K^t\| = e^{-t \operatorname{Tr} A/2}$;
- (c) if $D_t \geq 0$ fails, a witness exists whose Fourier transform is real, even and supported in a cone.

Lines 428–431.

Content. $K^t f \in H_C$ if and only if the left side of (3.4) is finite.

Why it holds. $K^t f$ is real, continuous and in L^2 by Lemma 3.2 and (3.1). Its transform is Hermitian because it is real. Now apply Lemma 2.1(b).

Lines 433–434.

Content. Sufficiency and the upper bound.

Why it holds. $D_t \geq 0$ gives $e^{-\frac{1}{4}\eta^\top D_t \eta} \leq 1$. So (3.4) and (2.5) give $\|K^t f\|_C^2 \leq e^{-t \operatorname{Tr} A} \|f\|_C^2$.

Lines 436–444.

Content. The lower bound, $\|K^t\|^2 \geq e^{-t \operatorname{Tr} A}$.

Why it holds. $F_\varepsilon = \mathbf{1}_{\{|\eta| < \varepsilon\}}$ is real and even, and $\int F_\varepsilon w_C < \infty$. So $f_\varepsilon = F_\varepsilon^\vee \in H_C \setminus \{0\}$. On the ball, $\eta^\top D_t \eta \leq \|D_t\| \varepsilon^2$, so

$$\|K^t f_\varepsilon\|_C^2 = e^{-t \operatorname{Tr} A} (2\pi)^{-d} \kappa_C^{-1} \int_{|\eta| < \varepsilon} w_C e^{-\frac{1}{4}\eta^\top D_t \eta} \geq e^{-t \operatorname{Tr} A} e^{-\frac{1}{4}\varepsilon^2 \|D_t\|} \|f_\varepsilon\|_C^2.$$

Let $\varepsilon \downarrow 0$.

Notes. The norm is attained asymptotically by low-frequency functions. There $e^{-\frac{1}{4}\eta^\top D_t \eta} \approx 1$, and only the Jacobian factor $e^{-t \operatorname{Tr} A}$ of the dilation remains.

Lines 446–450.

Content. Necessity and the witness.

Why it holds. Apply Lemma 2.2 with $S = -D_t$. Then $S \leq 0$ fails, and the resulting F_S gives $\int |F_S|^2 w_C e^{-\frac{1}{4}\eta^\top D_t \eta} = \infty$. The function $f = F_S^\vee$ lies in H_C with $\hat{f} = F_S$, and (3.4) shows that the left side is ∞ , so $K^t f \notin H_C$.

Verified. A.

Lines 452–461.

Content. Theorem 4.2: K^t maps H_C bijectively onto $H_{\tilde{C}_t}$, and $\|K^t f\|_{\tilde{C}_t}^2 = \tilde{\tau}_t \|f\|_C^2$. In other words, $\tilde{\tau}_t^{-1/2} K^t: H_C \rightarrow H_{\tilde{C}_t}$ is unitary.

Why it holds. $\tilde{C}_t$ is well defined and positive definite, because $C^2 + 4\Sigma(t) \geq C^2 > 0$ and congruence preserves positivity.

Lines 463–478.

Content. Image and isometry.

Why it holds. By the computation of Proposition 3.3 with weight $w_{\tilde{C}_t}$, the exponent is $-\eta^\top \Sigma \eta + \frac{1}{4}\eta^\top e^{At} \tilde{C}_t^2 e^{A^\top t} \eta$. Since $e^{At} \tilde{C}_t^2 e^{A^\top t} = C^2 + 4\Sigma(t)$, it equals

$$-\eta^\top \Sigma \eta + \frac{1}{4}\eta^\top (C^2 + 4\Sigma) \eta = \frac{1}{4}\eta^\top C^2 \eta.$$

So $\int |\hat{g}|^2 w_{\tilde{C}_t} = e^{-t \operatorname{Tr} A} \int |\hat{f}|^2 w_C$. Moreover $\det \tilde{C}_t = e^{-t \operatorname{Tr} A} \det(C^2 + 4\Sigma)^{1/2}$, hence

$$\|g\|_{\tilde{C}_t}^2 = \frac{\kappa_C}{\kappa_{\tilde{C}_t}} e^{-t \operatorname{Tr} A} \|f\|_C^2 = \frac{\det C e^{-t \operatorname{Tr} A}}{e^{-t \operatorname{Tr} A} \det(C^2 + 4\Sigma)^{1/2}} \|f\|_C^2 = \tilde{\tau}_t \|f\|_C^2.$$

Verified. A; S: S5 (the exponent identity, generic symmetric 2×2 matrices) and S2 (for $d = 1$ and $f = k_z^C$, $\|K^t k_z^C\|_{\tilde{C}_t}^2 = c/\sqrt{c^2 + 4\sigma} = \tilde{\tau}_t$, computed from an explicit Fourier integral). N: item 4 (Gram matrices, to 3.8×10^{-14}).

Lines 480–496.

Content. Surjectivity.

Why it holds. Put $F(\eta) = e^{t \operatorname{Tr} A} \hat{g}(e^{A^\top t} \eta) e^{\frac{1}{2} \eta^\top \Sigma \eta}$. It is Hermitian: $\hat{g}$ is Hermitian and the other factors are real and even. With $\eta = e^{-A^\top t} \xi$ and $d\eta = e^{-t \operatorname{Tr} A} d\xi$,

$$\eta^\top \Sigma \eta + \frac{1}{4} \eta^\top C^2 \eta = \frac{1}{4} \eta^\top (C^2 + 4\Sigma) \eta = \frac{1}{4} \xi^\top \tilde{C}_t^2 \xi.$$

So $\int |F|^2 w_C = e^{t \operatorname{Tr} A} \int |\hat{g}|^2 w_{\tilde{C}_t} < \infty$. By (3.2),

$$\widehat{K^t F^\vee}(\xi) = e^{-t \operatorname{Tr} A} \cdot e^{t \operatorname{Tr} A} \hat{g}(\xi) e^{\frac{1}{2} \xi^\top e^{-At} \Sigma e^{-A^\top t} \xi} \cdot e^{-\frac{1}{2} \xi^\top e^{-At} \Sigma e^{-A^\top t} \xi} = \hat{g}(\xi),$$

and continuity gives $K^t F^\vee = g$.

Verified. A.

Lines 498–509.

Content. Corollary 4.3. $K^t H_C \subset H_C \iff H_{\tilde{C}_t} \subset H_C \iff \tilde{C}_t^2 \geq C^2 \iff D_t \geq 0$. Moreover $H_{\tilde{C}_t} \subsetneq H_{C_t}$ when $\Sigma(t) \neq 0$, and the implication discussed in [PSWPN, Remark 4.2] fails for the matrices of Corollary 1.2.

Lines 511–516.

Content. The chain of equivalences, and strictness.

Why it holds. $\tilde{C}_t^2 - C^2 = e^{-At}(C^2 + 4\Sigma - e^{At} C^2 e^{A^\top t}) e^{-A^\top t} = e^{-At} D_t e^{-A^\top t}$, a congruence. Next, $\tilde{C}_t^2 - C_t^2 = e^{-At}(4\Sigma - 2\Sigma) e^{-A^\top t} = 2e^{-At} \Sigma e^{-A^\top t} \geq 0$. If both inclusions held, Corollary 2.3 would give $C_t^2 \geq \tilde{C}_t^2$ as well, so $\Sigma(t) \geq 0$ and $-\Sigma(t) \geq 0$, hence $\Sigma(t) = 0$. If $B \neq 0$ and $t > 0$, then $\operatorname{Tr} \Sigma(t) = \int_0^t \|B^\top e^{A^\top s}\|_F^2 ds > 0$, because the continuous integrand equals $\|B\|_F^2 > 0$ at $s = 0$. (Here $\|\cdot\|_F$ is the Frobenius norm.)

Verified. A; S: S5 (the identity $\tilde{C}_t^2 - C_t^2 = 2e^{-At} \Sigma e^{-A^\top t}$) and S3 (the congruence, Example 6.1).

Lines 518–524.

Content. The implication “ $K^t H_C \subset H_C \Rightarrow H_{C_t} \subset H_C$ ” fails for the matrices of Corollary 1.2.

Why it holds. $f_x(t) = \langle C_t^2 x, x \rangle$ with $C_t^2 = e^{-At}(C^2 + 2\Sigma(t))e^{-A^\top t}$. Using $\Sigma(0) = 0$ and $\dot{\Sigma}(0) = BB^\top$,

$$\begin{aligned} \left. \frac{d}{dt} C_t^2 \right|_{t=0} &= -AC^2 - C^2 A^\top + 2BB^\top, \\ \dot{f}_x(0) &= \langle (2BB^\top - (AC^2 + C^2 A^\top))x, x \rangle. \end{aligned}$$

For $\beta = 5$ the matrix $2I - (A + A^\top) = \begin{pmatrix} 4 & -5 \\ -5 & 4 \end{pmatrix}$ has the eigenvalue -1 .

So $\dot{f}_x(0) < 0$ for an eigenvector x , and $f_x(t) < f_x(0) = |Cx|^2$ on some $(0, \varepsilon)$. Corollary 2.3 then gives $H_{C_t} \not\subset H_C$. Invariance for all t comes from Theorem 4.4; there is no circularity, because Theorem 4.4 does not use Corollary 4.3.

Verified. A; N: item 6. For $\beta \in \{5, 6\}$, $C_t^2 - C^2$ has the negative eigenvalues -1.2×10^{-2} and -2.4×10^{-2} at some $t \leq 0.0124$. These match the first-order prediction $\lambda_{\min} \approx -t$ (resp. $-2t$) at $t = 0.0124$.

Notes. Correction C3 applies here.

Lines 526–533.

Content. Theorem 4.4. The following are equivalent: (i) invariance for all t ; (ii) invariance along a sequence $t_n \downarrow 0$; (iii) $\Lambda_C \geq 0$.

Lines 535–549.

Content. The proof via (4.2), $D_t = \int_0^t e^{As} \Lambda_C e^{A^\top s} ds$.

Why it holds. Clearly $D_0 = 0$, because $\Sigma(0) = 0$. By the fundamental theorem of calculus, $\frac{d}{dt} 4\Sigma(t) = 4e^{At} B B^\top e^{A^\top t}$. By the product rule and $Ae^{At} = e^{At} A$, $\frac{d}{dt}(e^{At} C^2 e^{A^\top t}) = e^{At}(AC^2 + C^2 A^\top)e^{A^\top t}$. So $\dot{D}_t = e^{At} \Lambda_C e^{A^\top t}$, and integrating gives (4.2).

- (iii) $\Rightarrow$ (i): the integrand in (4.2) is PSD.
- (ii) $\Rightarrow$ (iii): $t_n^{-1} D_{t_n} \geq 0$, and $t^{-1} D_t \rightarrow \Lambda_C$ by continuity of the integrand. The PSD cone is closed.

Verified. A; S: S3 checks (4.2), $\dot{D}_t$ and $\lim_{t \rightarrow 0} D_t/t = \Lambda_C$ for Example 6.1. N: item 2, for three random non-Hurwitz triples.

Lines 551–556.

Content. Remark 4.5. The same computation as in [PSWPN], with $\tilde{C}_t$ in place of C_t , gives $\tilde{f}_x(t) = \langle \tilde{C}_t^2 x, x \rangle = |C e^{-A^\top t} x|^2 + 4 \int_0^t |B^\top e^{-A^\top s} x|^2 ds$ and $\tilde{f}_x(t) = \langle \Lambda_C e^{-A^\top t} x, e^{-A^\top t} x \rangle$.

Why it holds. Put $y = e^{-A^\top t} x$. Then $\langle \tilde{C}_t^2 x, x \rangle = \langle (C^2 + 4\Sigma(t))y, y \rangle$. For the Σ term, substitute $r = t - s$:

$$\langle \Sigma(t)y, y \rangle = \int_0^t |B^\top e^{A^\top s} e^{-A^\top t} x|^2 ds = \int_0^t |B^\top e^{-A^\top r} x|^2 dr.$$

Differentiating,

$$\begin{aligned} \frac{d}{dt} |C e^{-A^\top t} x|^2 &= -\langle (AC^2 + C^2 A^\top)y, y \rangle, \\ \frac{d}{dt} 4 \int_0^t |B^\top e^{-A^\top r} x|^2 dr &= 4 \langle B B^\top y, y \rangle. \end{aligned}$$

Together this is $\langle \Lambda_C y, y \rangle$.

Verified. A; S: S3 checks $\frac{d}{dt} \tilde{C}_t^2 = e^{-At} \Lambda_C e^{-A^\top t}$ for Example 6.1.

Lines 558–563.

Content. The proof of Theorem 1.1.

Why it holds. (a) is Theorem 4.1. (b) is Theorem 4.4, (i) $\iff$ (iii). For the interval statement: if every interval $(0, 1/n)$ contained an admissible t_n , then $t_n \rightarrow 0$, and a strictly decreasing subsequence would satisfy (ii).

Lines 565–572.

Content. The proof of Corollary 1.2.

Why it holds. $BB^\top \geq 0$ gives $BB^\top \leq 2BB^\top$, so (1.3) implies (1.5). For the example, $A + A^\top = \begin{pmatrix} -2 & 5 \\ 5 & -2 \end{pmatrix}$ has the eigenvalues -2 ± 5 , so $\frac{1}{2}(A + A^\top)$ has the eigenvalues $\frac{3}{2}$ and $-\frac{7}{2}$. Hence (1.3) reads $\frac{3}{2} \leq 1$, which is false, and (1.5) reads $\frac{3}{2} \leq 2$, which is true. Moreover A is Hurwitz (double eigenvalue -1), $B = I$ is invertible (so (A, B) is controllable), and $m = d = 2$.

Verified. A; S: S3.

Lines 574–585.

Content. Corollary 4.6, the isotropic case: invariance for all t if and only if $\frac{1}{2}(A + A^\top) \leq \frac{2}{\sigma^2}BB^\top$. This sharpens [PSWPN, Corollary 2.3], which has $\frac{1}{\sigma^2}$.

Why it holds. Put $C = \sigma I$ in (1.5) and divide by σ^2 .

Verified. L: [PSWPN, Corollary 2.3] reads “assume that $\frac{1}{2}(A + A^\top) \leq \frac{1}{\sigma^2}BB^\top$ (e.g., if A is dissipative)”.

8. PAPER SECTION 5: STABLE DRIFT (LINES 588–642)

Lines 591–611.

Content. Theorem 5.1, for Hurwitz A and without controllability. Here $P_C = C^2 + 4\Sigma > 0$, and $\mathcal{E}_C = \{x : x^\top P_C^{-1}x \leq 1\}$.

- (a) $D_t = P_C - e^{At}P_Ce^{A^\top t}$, so invariance at time t is equivalent to $e^{At}\mathcal{E}_C \subset \mathcal{E}_C$.
- (b) $AP_C + P_CA^\top = -\Lambda_C$, so (1.5) is equivalent to forward invariance of $\mathcal{E}_C$; likewise for (1.3) with $C^2 + 2\Sigma$.
- (c) T_C is a closed additive semigroup, contains $[t_C, \infty)$, and obeys the dichotomy.

Why it holds. $P_C \geq C^2 > 0$.

Lines 613–619.

Content. Proof of (a).

Why it holds. First,

$$\int_t^\infty e^{As}BB^\top e^{A^\top s}ds = e^{At}\left(\int_0^\infty e^{Ar}BB^\top e^{A^\top r}dr\right)e^{A^\top t} = e^{At}\Sigma e^{A^\top t},$$

so $\Sigma(t) = \Sigma - e^{At}\Sigma e^{A^\top t}$. Insert this into D_t :

$$D_t = C^2 + 4\Sigma - 4e^{At}\Sigma e^{A^\top t} - e^{At}C^2 e^{A^\top t} = P_C - e^{At}P_C e^{A^\top t}.$$

With $G = P^{-1/2}EP^{1/2}$ we have $GG^\top = P^{-1/2}EPE^\top P^{-1/2}$. Therefore

$$EPE^\top \leq P \iff GG^\top \leq I \iff |G^\top z| \leq |z| \forall z \iff \|G\| = \|G^\top\| \leq 1.$$

Finally, $|P^{-1/2}Ex| = |Ex|_P$ and $|P^{-1/2}x| = |x|_P$. By homogeneity, contraction in a norm is the same as mapping its unit ball into itself.

Verified. A; S: S3 (Example 6.1). N: item 3 (random Hurwitz A , entries of the difference at most 2.4×10^{-14}).

Lines 621–626.

Content. Proof of (b).

Why it holds. $A\Sigma + \Sigma A^\top = \int_0^\infty \frac{d}{ds}(e^{As}BB^\top e^{A^\top s}) ds = 0 - BB^\top$, because $e^{As} \rightarrow 0$. Hence $AP_C + P_C A^\top = AC^2 + C^2 A^\top - 4BB^\top = -\Lambda_C$, and similarly with 2Σ . Forward invariance means $e^{At}P_C e^{A^\top t} \leq P_C$ for all t , by (a).

- If $AP_C + P_C A^\top \leq 0$, then $\frac{d}{dt}e^{At}P_C e^{A^\top t} = e^{At}(AP_C + P_C A^\top)e^{A^\top t} \leq 0$.
- Conversely, divide $P_C - e^{At}P_C e^{A^\top t} \geq 0$ by t and let $t \downarrow 0$.

Verified. A; S: S3 and S4 ($AP_C + P_C A^\top = -\Lambda_C$ exactly). N: item 3.

Lines 628–633.

Content. Proof of (c).

Why it holds. *Closed and contains 0:* $D_0 = 0$, $t \mapsto D_t$ is continuous, and the PSD cone is closed.

Additive: congruence by e^{As} preserves $\leq$, so $e^{A(s+t)}P_C e^{A^\top(s+t)} \leq e^{As}P_C e^{A^\top s} \leq P_C$.

Tail: $x^\top e^{At}P_C e^{A^\top t}x \leq \lambda_{\max}(P_C)|e^{A^\top t}x|^2 \leq \lambda_{\max}(P_C)\|e^{At}\|^2|x|^2$. This is at most $\lambda_{\min}(P_C)|x|^2 \leq x^\top P_C x$ when $\|e^{At}\|^2 \leq \lambda_{\min}/\lambda_{\max}$. Such t exist because $\|e^{At}\| \rightarrow 0$.

Dichotomy: Theorem 1.1(b).

Verified. A.

Notes. T_C need not be an interval in general. The paper claims only the semigroup and tail structure, and the interval shape in Example 6.1 is labelled as numerical.

Lines 635–642.

Content. Remark 5.2. The ellipsoid of Σ is forward invariant whenever $\Sigma > 0$. Invariance of H_C is forward invariance of the ellipsoid of $\Sigma + \frac{1}{4}C^2$; condition (1.3) asks this for $\Sigma + \frac{1}{2}C^2$. For Hurwitz A , every H_C is invariant for large t .

Why it holds. $\frac{d}{dt}e^{At}\Sigma e^{A^\top t} = -e^{At}BB^\top e^{A^\top t} \leq 0$. The ellipsoids of P and of cP ($c > 0$) are dilates of each other, so forward invariance does not depend

on c ; here $P_C = 4(\Sigma + \frac{1}{4}C^2)$ and $C^2 + 2\Sigma = 2(\Sigma + \frac{1}{2}C^2)$. The last sentence is Theorem 5.1(c).

Notes. Correction C5 applies here.

9. PAPER SECTION 6: EXAMPLES (LINES 645–732)

Lines 648–661.

Content. Example 6.1: $A_\beta = \begin{pmatrix} -1 & \beta \\ 0 & -1 \end{pmatrix}$, $B = C = I_2$, $\beta \geq 0$. Condition (1.3) holds if and only if $\beta \leq 4$, and (1.5) if and only if $\beta \leq 6$. For $4 < \beta \leq 6$ the space is invariant with $\|K^t\| = e^t$; for $\beta > 6$ invariance fails for small t .

Why it holds. $A + A^\top = \begin{pmatrix} -2 & \beta \\ \beta & -2 \end{pmatrix}$ has the eigenvalues $-2 \pm \beta$. For $\beta \geq 0$ the largest eigenvalue of $\frac{1}{2}(A + A^\top)$ is $\frac{\beta-2}{2}$. So (1.3) reads $\frac{\beta-2}{2} \leq 1$, i.e. $\beta \leq 4$, and (1.5) reads $\frac{\beta-2}{2} \leq 2$, i.e. $\beta \leq 6$. Also $\text{Tr } A_\beta = -2$ gives $e^{-t \text{Tr } A/2} = e^t$.

Verified. A; S: S3 (eigenvalues, both thresholds, trace). N: item 6 ($\beta \in \{4, \dots, 8\}$).

Lines 663–668.

Content. $e^{A_\beta t}$ and the integrand of $\Sigma(t)$.

Why it holds. $A_\beta = -I + \beta E_{12}$, and $-I$ commutes with E_{12} while $E_{12}^2 = 0$. So $e^{A_\beta t} = e^{-t}(I + \beta t E_{12})$, and

$$e^{A_\beta s} e^{A_\beta^\top s} = e^{-2s} \begin{pmatrix} 1 & \beta s \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \beta s & 1 \end{pmatrix} = e^{-2s} \begin{pmatrix} 1 + \beta^2 s^2 & \beta s \\ \beta s & 1 \end{pmatrix}.$$

Verified. A; S: S3 (SymPy's matrix exponential agrees).

Lines 669–678.

Content. Three elementary integrals and the entries of D_t .

Why it holds. *Antiderivatives.* The antiderivatives of e^{-2s} , se^{-2s} and $s^2 e^{-2s}$ are $-\frac{1}{2}e^{-2s}$, $-e^{-2s}(\frac{s}{2} + \frac{1}{4})$ and $-e^{-2s}(\frac{s^2}{2} + \frac{s}{2} + \frac{1}{4})$. For the last one, differentiate: $e^{-2s}(s^2 + s + \frac{1}{2}) - e^{-2s}(s + \frac{1}{2}) = s^2 e^{-2s}$. Evaluating from 0 to t gives the three formulas.

The entries. Write $e = e^{-2t}$. Then $4\Sigma_{11} = 2(1-e) + \beta^2(1-e(1+2t+2t^2))$, and $(e^{At} e^{A^\top t})_{11} = e(1 + \beta^2 t^2)$. Hence

$$\begin{aligned} d_{11} &= 1 + 2(1-e) + \beta^2(1-e(1+2t+2t^2)) - e(1 + \beta^2 t^2) \\ &= 3(1-e) + \beta^2(1-e(1+2t+3t^2)). \end{aligned}$$

Similarly $d_{12} = \beta(1-e(1+2t)) - e\beta t = \beta(1-e(1+3t))$ and $d_{22} = 1 + 2(1-e) - e = 3(1-e)$.

Verified. A; S: S3 (the three integrals and the closed form of D_t , exactly). N: item 6 (closed form against numerical D_t to 2.5×10^{-14}).

Lines 679–683.

Content. $D_t = t\Lambda_C + O(t^2)$ with $\Lambda_C = \begin{pmatrix} 6 & -\beta \\ -\beta & 6 \end{pmatrix}$.

Why it holds. $\Lambda_C = 4I - (A + A^\top)$. Also directly: $1 - e^{-2t} = 2t + O(t^2)$, $1 - e^{-2t}(1 + 3t) = -t + O(t^2)$ and $1 - e^{-2t}(1 + 2t + 3t^2) = O(t^2)$.

Verified. S: S3 ($\lim_{t \rightarrow 0} D_t/t = \Lambda_C$).

Lines 684–694.

Content. Σ , P_C , and the two Lyapunov matrices, which recover the thresholds 6 and 4.

Why it holds. Let $t \rightarrow \infty$ in the integrals: $\Sigma_{11} = \frac{1}{2} + \frac{\beta^2}{4}$, $\Sigma_{12} = \frac{\beta}{4}$, $\Sigma_{22} = \frac{1}{2}$.

So $P_C = I + 4\Sigma = \begin{pmatrix} 3 + \beta^2 & \beta \\ \beta & 3 \end{pmatrix}$. Then $A_\beta P_C = \begin{pmatrix} -3 & 2\beta \\ -\beta & -3 \end{pmatrix}$, and adding

the transpose gives $\begin{pmatrix} -6 & \beta \\ \beta & -6 \end{pmatrix}$. This is negative semidefinite if and only

if $\beta \leq 6$. Likewise $A_\beta(I + 2\Sigma) = \begin{pmatrix} -2 & \frac{3}{2}\beta \\ -\frac{1}{2}\beta & -2 \end{pmatrix}$, and with its transpose

$\begin{pmatrix} -4 & \beta \\ \beta & -4 \end{pmatrix}$, which is negative semidefinite if and only if $\beta \leq 4$.

Verified. A; S: S3 (all four matrices; $A\Sigma + \Sigma A^\top = -I$; Theorem 5.1(a) identity).

Lines 696–703.

Content. The numerically computed boundary $t_1(\beta)$ for $6 < \beta \leq 14$, and a rigorous argument for $t \geq 40$.

Why it holds. Numerical part. For each of 320 values of β , `make_figure.py` samples $\lambda_{\min}(D_t)$ on 2×10^5 points of $(0, 40]$. It asserts that the admissible grid points form a final segment, and bisects on $\det D_t$. The results are $t_1(7) = 0.300056$ and $t_1(8) = 0.387611$.

Rigorous tail. $\|e^{A_\beta t}\| \leq e^{-t}(1 + \beta t)$, and $\frac{d}{dt}(1 + \beta t)e^{-t} = e^{-t}(\beta - 1 - \beta t) < 0$ for $t \geq 1$. At $t = 40$ and $\beta = 14$, $(1 + 560)e^{-40} = 2.38 \times 10^{-15} < 10^{-14}$.

For $P_C = \begin{pmatrix} 3 + \beta^2 & \beta \\ \beta & 3 \end{pmatrix}$, the ratio $r = \lambda_{\min}/\lambda_{\max}$ satisfies $r/(1 + r)^2 = \det P_C/(\text{Tr } P_C)^2 = (9 + 2\beta^2)/(6 + \beta^2)^2$. This function decreases in $\beta > 0$: its derivative is $-4\beta(\beta^2 + 3)/(\beta^2 + 6)^3$. Since $r \mapsto r/(1 + r)^2$ increases on $(0, 1)$, r also decreases in β . At $\beta = 14$ the eigenvalues are $101 \mp 70\sqrt{2}$, and

$$\frac{101 - 70\sqrt{2}}{101 + 70\sqrt{2}} \geq \frac{1}{100} \iff 9999 \geq 7070\sqrt{2}$$

$$\iff 9999^2 = 99980001 \geq 2 \cdot 7070^2 = 99969800,$$

so $r \geq 10^{-2}$ for $\beta \leq 14$. Then $\lambda_{\max}\|e^{At}\|^2 < \lambda_{\min}$ for $t \geq 40$, and the proof of Theorem 5.1(c) gives $D_t > 0$ strictly.

Verified. S: S3 checks the monotonicity, the exact eigenvalues, the inequality $9999^2 \geq 2 \cdot 7070^2$ and the bound $561e^{-40} < 10^{-14}$ with 50 digits. N: item 7 and `figure_output.txt`.

Notes. The ratio at $\beta = 14$ is 0.010026, so the margin over 10^{-2} is small but exact. The margin in the other factor is enormous (10^{-28} against 10^{-2}). The conclusion “ $T_I = \{0\} \cup [t_1(\beta), \infty)$ ” is stated “up to the grid resolution”, which is accurate.

Lines 706–716.

Content. Figure 1: the admissible region in the (β, t) plane. The region $\beta \leq 6$ is rigorous (Theorem 4.4), the hatched region $\beta \leq 4$ is covered by (1.3), and the curve $t_1(\beta)$ for $\beta > 6$ is numerical.

Verified. N: `make_figure.py`; the legend references “(1.3)” and “Theorem 4.1” match the compiled numbering. The fonts are embedded as TrueType, not Type 3.

Lines 718–732.

Content. Example 6.2, scalar noise: $A = \begin{pmatrix} \frac{3}{2} & -2 \\ 2 & -2 \end{pmatrix}$, $B = e_1$, $C = I$. Here (1.3) fails and (1.5) holds.

Why it holds. $\text{Tr } A = -\frac{1}{2}$ and $\det A = -3 + 4 = 1$, so $\lambda^2 + \frac{1}{2}\lambda + 1 = 0$, i.e. $\lambda = -\frac{1}{4} \pm \frac{\sqrt{15}}{4}i$. Next, $AB = (\frac{3}{2}, 2)^\top$ and $\det[B, AB] = \det \begin{pmatrix} 1 & \frac{3}{2} \\ 0 & 2 \end{pmatrix} = 2$. Further $\frac{1}{2}(A + A^\top) = \text{diag}(\frac{3}{2}, -2)$ and $BB^\top = \text{diag}(1, 0)$.

Notes. The paper does not record the following, which gives an independent confirmation through Theorem 5.1(b). The stationary covariance is $\Sigma = \begin{pmatrix} 5 & 4 \\ 4 & 4 \end{pmatrix} > 0$. Indeed $A\Sigma = \begin{pmatrix} -\frac{1}{2} & -2 \\ 2 & 0 \end{pmatrix}$, so $A\Sigma + \Sigma A^\top = \text{diag}(-1, 0) = -BB^\top$. Then $P_C = \begin{pmatrix} 21 & 16 \\ 16 & 17 \end{pmatrix}$ and $AP_C + P_C A^\top = \text{diag}(-1, -4) \leq 0$.

Verified. A; S: S4 (all of the above, exactly). N: item 8 ($\lambda_{\min}(D_t) \geq 0$ on 3000 grid points).

10. PAPER SECTION 7: FURTHER REMARKS (LINES 735–863)

Lines 738–755.

Content. Lemma 7.1. $k_y^{C'} \in H_{C''}$ if and only if $G = 2C'^2 - C''^2 > 0$, and then

$$\langle k_y^{C'}, k_{y'}^{C'} \rangle_{C''} = \frac{(\det C')^2}{\det C''(\det G)^{1/2}} e^{-(y-y')^\top G^{-1}(y-y')}.$$

Why it holds. By (2.3), $|\widehat{k_y^{C'}}|^2 w_{C''} = \kappa_{C'}^2 e^{-\frac{1}{2}\xi^\top C'^2 \xi + \frac{1}{4}\xi^\top C''^2 \xi} = \kappa_{C'}^2 e^{-\frac{1}{4}\xi^\top G \xi}$. Diagonalizing G shows that $\int e^{-\frac{1}{4}\xi^\top G \xi} d\xi$ factorizes into one-dimensional integrals, which are all finite if and only if every eigenvalue of G is positive.

Next, (2.1) with $P = \frac{1}{4}G$ gives $\pi^{d/2}2^d(\det G)^{-1/2}e^{-(y-y')^\top G^{-1}(y-y')}$. The constants combine to

$$(2\pi)^{-d} \frac{\pi^d (\det C')^2}{\pi^{d/2} \det C''} \pi^{d/2} 2^d (\det G)^{-1/2} = \frac{(\det C')^2}{\det C'' (\det G)^{1/2}}.$$

For $C' = C'' = C$ we get $G = C^2$ and recover $k^C(y, y')$, a consistency check. *Verified.* A; S: S1 (the constant for general d ; the full formula for $d = 1$).

Lines 757–772.

Content. Remark 7.2.

- The kernel-section formula, derived from (3.2) and (2.3).
- $K^t k_z^C \in H_C \iff 2C_t^2 - C^2 > 0$.
- For small t , finite kernel expansions are mapped into H_C whatever the sign of D_t .

Why it holds. The derivation is the one given for lines 148–160 in Section 4. Lemma 7.1 with $C' = C_t$ and $C'' = C$ gives the membership criterion. Next,

$$2C_t^2 - C^2 = e^{-At}(2C^2 + 4\Sigma - e^{At}C^2e^{A^\top t})e^{-A^\top t} = e^{-At}(C^2 + D_t)e^{-A^\top t}.$$

Since $D_t \rightarrow 0$ and $C^2 > 0$, this is positive definite for small t .

For a finite expansion $f = \sum_j \alpha_j k_{y_j}^C$, (2.3) gives $|\hat{f}| \leq \kappa_C(\sum_j |\alpha_j|)w_C^{-1}$. So $|\hat{f}|^2 w_C \leq c w_C^{-1}$, and

$$e^{-t \operatorname{Tr} A} \int |\hat{f}|^2 w_C e^{-\frac{1}{4}\eta^\top D_t \eta} \leq c e^{-t \operatorname{Tr} A} \int e^{-\frac{1}{4}\eta^\top (C^2 + D_t) \eta} < \infty$$

whenever $C^2 + D_t > 0$.

Verified. A; L: [PSWPN, Remark 4.2] reads “Note that $K^t H_C \subset H_C$ at least implies $H_{0,C_t} = \operatorname{span}\{k_x^{C_t} : x \in \mathbb{R}^d\} \subset H_C$, see Proposition 4.1.”

Notes. Corrections C2 and C4 apply here.

Lines 774–791.

Content. Remark 7.3.

- Theorem 4.2 refines [PSWPN, Proposition 4.1]: it recovers the bound $\|K^t f\|_{\tilde{C}_t}^2 \leq \tau_t \|f\|_C^2$.
- On kernel sections, $\langle K^t k_z^C, K^t k_{z'}^C \rangle_{\tilde{C}_t} = \tilde{\tau}_t k^C(z, z')$.

Why it holds. *The bound.* Corollary 2.3 with $C_1 = \tilde{C}_t$ and $C_2 = C_t$ gives the factor $\det \tilde{C}_t / \det C_t$. Then

$$\frac{\det \tilde{C}_t}{\det C_t} \tilde{\tau}_t = \frac{e^{-t \operatorname{Tr} A} \det(C^2 + 4\Sigma)^{1/2}}{e^{-t \operatorname{Tr} A} \det(C^2 + 2\Sigma)^{1/2}} \cdot \frac{\det C}{\det(C^2 + 4\Sigma)^{1/2}} = \tau_t.$$

The kernel identity. Apply Lemma 7.1 with $C' = C_t$, $C'' = \tilde{C}_t$ and $G = 2C_t^2 - \tilde{C}_t^2 = e^{-At}C^2e^{-A^\top t}$. Then $G^{-1} = e^{A^\top t}C^{-2}e^{At}$, and the exponent becomes $(z - z')^\top C^{-2}(z - z')$. Let $a = \det(C^2 + 2\Sigma)$, $b = \det(C^2 + 4\Sigma)$,

$c = \det C$ and $e = e^{t \operatorname{Tr} A}$. Then $\tau_t^2 = c^2/a$, $(\det C_t)^2 = a/e^2$, $\det \tilde{C}_t = \sqrt{b}/e$ and $(\det G)^{1/2} = c/e$. So the constant is

$$\frac{c^2}{a} \cdot \frac{a/e^2}{(\sqrt{b}/e)(c/e)} = \frac{c}{\sqrt{b}} = \tilde{\tau}_t.$$

Verified. A; S: S5 checks both determinant identities and the formula for $2C_t^2 - \tilde{C}_t^2$. N: item 4 (Gram matrices). L: the statement of [PSWPN, Proposition 4.1] gives the norm bound $\tau_t^{1/2}$.

Lines 793–800.

Content. Remark 7.4, for $B = 0$. The composition operator $f \mapsto f \circ E$ maps H_C into itself if and only if $EC^2E^\top \leq C^2$, i.e. $\|C^{-1}EC\| \leq 1$, and then its norm is $|\det E|^{-1/2}$. The remark also cites [4] and the Fock-space analogue [2].

Why it holds. The dilation formula of Lemma 3.2 gives $\int |\widehat{f \circ E}|^2 w_C = |\det E|^{-1} \int |\widehat{f}|^2 w_C e^{-\frac{1}{4}\eta^\top (C^2 - EC^2E^\top)\eta}$. From here the proof of Theorem 4.1 runs verbatim, with $D = C^2 - EC^2E^\top$. Moreover $C^{-1}EC(C^{-1}EC)^\top = C^{-1}EC^2E^\top C^{-1}$, and $\|M\| \leq 1 \iff MM^\top \leq I$.

Verified. A; L: the abstract of [4] (arXiv:1911.11992) says that only affine maps induce bounded composition operators on a large class of RKHSs of analytic positive definite functions. The Fock-space statement is the main theorem of [2]; its bibliographic data were checked earlier (Section 12).

Notes. Translations $f \mapsto f(\cdot + b)$ are isometries of H_C , because they multiply $\widehat{f}$ by $e^{i\xi \cdot b}$. So every affine map $Ex + b$ with $\|C^{-1}EC\| \leq 1$ gives a bounded operator. This could be added in one sentence.

Lines 802–807.

Content. The Fourier description of H_φ for a translation-invariant kernel, in the paper's normalization.

Why it holds. This is the normalization conversion given for lines 286–288 in Section 5.

Lines 809–820.

Content. Proposition 7.5. $K^t H_\varphi \subset H_\varphi$ if and only if $M_t = \sup m_t < \infty$, and then $\|K^t\|^2 = e^{-t \operatorname{Tr} A} M_t$. For the Gaussian kernel, $m_t = e^{-\frac{1}{4}\eta^\top D_t \eta}$.

Why it holds. Gaussian case: $\widehat{\varphi} = \kappa_C w_C^{-1}$, so

$$m_t(\eta) = \exp\left(-\frac{1}{4}\eta^\top C^2 \eta - \eta^\top \Sigma \eta + \frac{1}{4}\eta^\top e^{At} C^2 e^{A^\top t} \eta\right) = e^{-\frac{1}{4}\eta^\top D_t \eta}.$$

Verified. S: S2 ($d = 1$). N: item 9 (1000 random η , relative error 3.2×10^{-14}).

Lines 822–832.

Content. The norm formula.

Why it holds. The substitution of Proposition 3.3 with weight $1/\widehat{\varphi}$ gives

$$\|K^t f\|^2 = (2\pi)^{-d} e^{-t \operatorname{Tr} A} \int |\widehat{f}|^2 m_t / \widehat{\varphi}.$$

- *Upper bound:* $m_t \leq M_t$.
- *Lower bound:* φ is real and positive definite, hence even, so $\widehat{\varphi}$ and m_t are even; and m_t is continuous, since $\widehat{\varphi}$ is continuous and positive. Choose η_0 with $m_t(\eta_0) > M_t - \delta$, where $0 < \delta < M_t$. For small r , $m_t > M_t - \delta$ on $U_r = B_r(\eta_0) \cup B_r(-\eta_0)$. The function $f_r = (\mathbf{1}_{U_r})^\vee$ lies in H_φ , because $\mathbf{1}_{U_r}$ is real and even and $1/\widehat{\varphi}$ is bounded on the compact set $\overline{U_r}$. It satisfies the stated inequality.

Verified. A.

Lines 834–842.

Content. If $M_t = \infty$, a witness exists.

Why it holds. The sets $\{m_t > 16^k\}$ are open, nonempty and symmetric. Choose bounded symmetric E_k of positive measure in them (for example two small balls), and put $a_k = (\int_{E_k} \widehat{\varphi}^{-1})^{-1/2}$. Then:

- each term of F has norm 2^{-k} in $L^2(\widehat{\varphi}^{-1} d\xi)$, so by Minkowski $\|F\| \leq \sum_{k \geq 1} 2^{-k} = 1$;
- $F \in L^1$ by Cauchy–Schwarz with $\int \widehat{\varphi} < \infty$;
- $F \in L^2$ because $0 < \widehat{\varphi} \leq \|\varphi\|_{L^1}$;
- on E_k , $m_t > 16^k$, so $\int F^2 m_t / \widehat{\varphi} \geq 4^{-k} a_k^2 16^k \int_{E_k} \widehat{\varphi}^{-1} = 4^k$ for every k .

Verified. A.

Lines 844–856.

Content. Example 7.6. For $\widehat{\varphi} = (1 + |\xi|^2)^{-s}$ with $s > d/2$ (Matérn/Sobolev), every K^t leaves H^s invariant, with $\|K^t\| \leq e^{-t \operatorname{Tr} A/2} \max\{1, \|e^{At}\|\}^s$.

Why it holds. $\widehat{\varphi}$ is integrable if and only if $2s > d$. Next, $(1 + a|\eta|^2)/(1 + |\eta|^2) \leq \max\{1, a\}$ with $a = \|e^{A^\top t}\|^2 = \|e^{At}\|^2$. With $e^{-\eta^\top \Sigma \eta} \leq 1$ this gives $m_t \leq \max\{1, \|e^{At}\|^2\}^s$, and the square root gives the norm bound.

Verified. A; N: item 10 (8.04×10^4 points, non-Hurwitz A).

Notes. Correction C6 applies here. For kernels whose spectral density decays like $e^{-|\xi|}$, the Gaussian factor $e^{-\eta^\top \Sigma(t) \eta}$ dominates once $\Sigma(t) > 0$. So the Lyapunov-type restriction is specific to spectral densities with (at least) Gaussian decay.

Lines 858–863.

Content. Remark 7.7 (Outlook): the method needs the “dilation times Gaussian” structure, which fails for nonlinear drifts or state-dependent noise.

11. APPENDIX A, THE AI-USE STATEMENT AND DATA AVAILABILITY
(LINES 865–939)

Lines 871–876.

Content. The three scripts. Errors are measured in $|M|_{\max}$. $\Sigma(t)$ is computed by Van Loan’s formula on steps of length at most 0.25, combined with the recursion $\Sigma(s+h) = \Sigma(h) + e^{Ah}\Sigma(s)e^{A^\top h}$.

Why it holds. The recursion is $\Sigma(s+h) = \int_0^h + \int_h^{s+h}$, and the substitution $u = r - h$ in the second integral gives $e^{Ah}\Sigma(s)e^{A^\top h}$. The one-step formula multiplies e^{-At} by e^{At} and loses all digits for large t ; this was observed during development and fixed.

Verified. N: item 1.

Lines 877–925.

Content. Items 1–10 and totals. Each number in the paper was compared with the script output of 24 September 2026.

Item	Claim in the paper	Script output	Match
1	$ \Sigma - \Sigma_q _{\max} \leq 2.7 \times 10^{-13} \Sigma_q _{\max}; 3 \times 3, 3 \times 2$	$\max 2.656 \times 10^{-13}$	yes
2	$\leq 5.2 \times 10^{-13} \max\{1, R_t _{\max}\};$ abscissae 1.645, 1.556, 2.043; 3.7×10^{-6} at $t = 10^{-6}$	$\max 5.177 \times 10^{-13};$ same abscissae; max 3.638×10^{-6}	yes
3	entries $\leq 2.4 \times 10^{-14}$ and $\leq 1.8 \times 10^{-14}$	$2.309 \times 10^{-14};$ 1.776×10^{-14}	yes
4	Gram entries within $3.8 \times 10^{-14};$ relative error 1.1×10^{-13}	$3.747 \times 10^{-14};$ 1.078×10^{-13}	yes
5	$0.436373 \pm 2.2 \times 10^{-4}$ vs. $0.436051; 1.45$ standard errors	same; ratio to 5 standard errors 0.2901	yes
6	(1.3) only for $\beta = 4; (1.5)$ for $\beta \leq 6; indefinite$ segments end at 0.298, 0.385; $-1.2 \times 10^{-2},$ $-2.4 \times 10^{-2}; 2.5 \times 10^{-14}$	as claimed; 0.2980, 0.3847; $-1.215 \times 10^{-2},$ $-2.415 \times 10^{-2};$ 2.487×10^{-14}	yes
7	320 values; interval; t_1 from 0.053 to 0.537	0.0526; 0.5366; assertions pass	yes
8	eigenvalues $-0.25 \pm 0.968i;$ $\lambda_{\max} = \pm 0.5; 3000$ points	same; $\min \lambda_{\min} = 10^{-4}$	yes
9	relative error 3.2×10^{-14}	3.197×10^{-14}	yes
10	8.04×10^4 points, $10^{-3} \leq \eta \leq 10^6,$ abscissa 0.857	$402 \text{ radii} \times 200 \text{ unit}$ directions; same abscissa	yes
Total	38 checks; 52 exact checks	38/38; 52/52	yes

Notes. Correction C10 applies here. Before the audit, item 2 wrote the spectral norm although the script uses the entrywise maximum. The parameters (dimensions, grids, seeds) were read off the code of `verify.py`. Its random generator is `numpy.random.default_rng(20260924)`.

Lines 927–935.

Content. A commented AI-use statement, to be activated and adapted by the author.

Notes. This is a blocking item; see Section 14. The statement names the tool (Claude, Anthropic), what it was used for, and the author’s responsibility. Its last sentence must be made true by the author before it is uncommented.

Lines 937–939.

Content. Data availability: no external data; the scripts are supplementary material.

12. BIBLIOGRAPHY (LINES 941–1019)

For each entry: what it supports, and how the bibliographic data were checked. “Web” means a web search in this or an earlier step of the session. “PDF” means the arXiv PDF of [PSWPN] supplied with the task.

- [1]: Aronszajn 1950 supports the Moore–Aronszajn uniqueness argument (lines 269–279). Bibliographic data: standard; web.
- [2]: Carswell–MacCluer–Schuster, Fock-space composition operators (Remark 7.4). Web.
- [3]: Hertel–Philipp–Schaller–Worthmann (new) supports lines 77 and 855. Web: arXiv:2512.20247, DOI 10.1137/26M1839689 [20]. *Caveat:* volume 25, issue 3 and pages 2044–2083 come from search metadata, because the SIAM page is blocked here. Confirm them at the DOI before submission.
- [4]: Ikeda–Ishikawa–Sawano (new), Remark 7.4. Web: *J. Math. Anal. Appl.* 511 (2022), no. 1, 126048; arXiv:1911.11992 [22].
- [5]: Klus–Schuster–Muandet, *J. Nonlinear Sci.* 30 (2020). Web.
- [6]: Köhne et al. (updated) supports lines 77 and 855. Web: DOI 10.1137/24M1650120, SIADS 24 (2025) 501–529 [21]. *Caveat:* the page range comes from search metadata; confirm it at the DOI.
- [7]: Koopman 1931. Web.
- [8]: Kostic et al., NeurIPS 35 (2022). Web.
- [9]: Mauroy–Mezić–Susuki (eds.), LNCIS vol. 484, Springer 2020. Web [23].
- [10]: Paulsen–Raghupathi 2016, Cambridge Studies in Advanced Mathematics 152. Web.
- [11]: Philipp et al., *Appl. Comput. Harmon. Anal.* 71 (2024) 101657. Web.

- [12]: [PSWPN], PAMM 25 (2025) e202400127 and arXiv:2405.14429. PDF for all numbered statements; web for the PAMM data [24]. The entry now notes that numbered statements follow arXiv v1. The PAMM version could not be opened (Wiley is blocked here), so its numbering may differ.
- [13]: Steinwart–Hush–Scovel, *IEEE Trans. Inform. Theory* 52 (2006). Web.
- [14]: Van Loan, *IEEE Trans. Automat. Control* 23 (1978). Web. This is the block-exponential method used in the scripts.
- [15]: Vatiwutipong–Phewchean, *Adv. Difference Equ.* 2019, Paper No. 276. Web.
- [16]: Wendland 2005, Cambridge Monographs 17. Web for the book. The content of Theorem 10.12 is from memory; see the note on lines 286–288.
- [17], [18]: Williams–Kevrekidis–Rowley, *J. Nonlinear Sci.* 25 (2015); Williams–Rowley–Kevrekidis, *J. Comput. Dyn.* 2 (2015). Web.

Every entry is cited in the text, and every citation has an entry (B: no undefined citations in the log). The keys are ordered so that citation lists appear in increasing numerical order.

13. QUESTIONS A REFEREE MAY ASK

- (1) *Is K^t well defined on H_C , and is it the operator of [PSWPN]?* Yes. $K^t f$ is defined pointwise by (3.1), and Remark 3.1 shows agreement with the $L^2(\mu)$ operator of [PSWPN] under their assumptions.
- (2) *Is the proof not just a Fourier computation?* The proof is short. Its value is threefold:
 - it settles a published question with a sharp answer (Theorem 1.1);
 - it identifies the range of K^t exactly (Theorem 4.2);
 - it pinpoints where the earlier argument loses information (Corollary 4.3, Remarks 7.2 and 7.3).
 A cover letter should say this plainly.
- (3) *What is the relation to the literature on Ornstein–Uhlenbeck semigroups?* The paper does not discuss it. A referee from probability may ask for a paragraph relating (3.2) to the Mehler formula and to the classical theory of Ornstein–Uhlenbeck semigroups on $L^p(\mu)$. Such a paragraph would need references that were not verified in this audit, so none were added.
- (4) *Do the standing assumptions matter?* No. Theorem 1.1 holds for all A , B and m . Corollary 1.2 and Examples 6.1 and 6.2 satisfy the standing assumptions, so the answer to [PSWPN] is given within their setting.

- (5) *Is T_C always an interval $\{0\} \cup [t_1, \infty)$?* The paper does not claim this. It proves the semigroup and tail structure (Theorem 5.1(c)) and reports the interval shape in Example 6.1 as numerical.
- (6) *What about nonlinear SDEs?* This is outside the scope, and Remark 7.7 says so.
- (7) *How does this relate to [3]?* That paper treats Sobolev spaces for stochastic dynamics. Example 7.6 is the elementary linear case, and the Gaussian question is not addressed there.

14. SUBMISSION READINESS AND CHOICE OF JOURNAL

14.1. **Verdict.** The *content* is ready:

- the mathematics is complete and every step was re-derived;
- the claims are checked symbolically and numerically;
- the literature is up to date as far as the searches reach;
- the manuscript compiles without warnings.

The *submission* is not ready, because of the blocking items below. None of them is mathematical.

14.2. **Blocking items.**

- (1) *Authorship.* Fill in the author block (lines 38–41): name, affiliation, e-mail and ORCID. The submitting author must be able to defend every line. SIAM’s editorial policy (version 2.0, May 2026) states that authors must be human and able to directly defend all aspects of the submission [26]. This commentary is meant to make that possible. If the author cannot follow a derivation here, a colleague in functional analysis or kernel methods should read the paper before submission.
- (2) *AI-use disclosure.* Activate and adapt lines 927–935 according to the journal’s policy.
 - SIAM requires disclosure of substantive AI use, and purposeful non-disclosure is treated as misconduct [26, 27].
 - Elsevier requires a separate AI declaration before the references [28].
 - For the AMS I could not find the current policy; check it before submitting there.
- (3) *Journal format and declarations.* Use the journal’s class: `siamart` for SIAM, `elsarticle` for Elsevier, the AMS `proc` class for PAMS. Add funding, competing-interest and code-availability statements as required.
- (4) *Cover letter.* It should name the open question, its source ([PSWPN, Remark 2.2(c), Remark 4.2]), the answer, and the three contributions listed in Section 13, item 2.

14.3. Recommended items.

- (1) Post the paper on arXiv (math.FA, cross-listed to math.PR and math.DS) before or at submission, to establish priority.
- (2) Repeat the novelty search on arXiv and MathSciNet/zbMATH, which could not be reached from here (Section 2.4).
- (3) Confirm the volume and pages of [3] and [6] at their DOIs, and Theorem 10.12 of [16] in the book.
- (4) Consider informing Philipp et al. as a courtesy. They posed the question and are natural referees.
- (5) Supply the three scripts and their outputs as supplementary material; the README lists them.
- (6) Optional additions: MSC 47B32; the observation that (1.5) is the condition of [PSWPN] for the squared kernel (Section 4); the remark on affine maps (Section 10).

14.4. Choice of journal. *First choice: SIAM Journal on Applied Dynamical Systems (SIADS).*

- Its scope includes contributions “to the theory of dynamical systems” [25].
- The error analysis of kernel EDMD that motivated the question is published there: Köhne et al. (2025) and Hertel et al. (2026). The readers who need the result read SIADS.
- SIAM’s policy permits AI-assisted research with disclosure and human accountability [26].
- The length (19 pages in `amsart`) is ordinary for SIADS.

Risk: an editor may find a purely theoretical note on linear SDEs narrow for SIADS. Section 5 (invariant ellipsoids, admissible times) and Figure 1 supply the dynamical-systems content; the cover letter should stress them.

Alternative 1: J. Math. Anal. Appl. (JMAA).

- The scope covers functional analysis and operator theory, dynamical systems and probability [29].
- JMAA published the closely related composition-operator result [4].
- It is a safe venue for a complete, correct result of this size.
- Elsevier’s AI declaration is required [28].

Alternative 2: Proceedings of the AMS. Papers may not exceed 15 printed pages [30]. The core (Sections 1–5 and Example 6.1) fits if Appendix A, Remarks 7.3 and 7.4 and Proposition 7.5 with Example 7.6 move to supplementary material. PAMS gives high visibility to short complete answers, but an editor may consider the topic specialized.

Not recommended: PAMM, where [PSWPN] appeared. It publishes contributions to the GAMM annual meetings [24], not independent research articles of this kind.

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